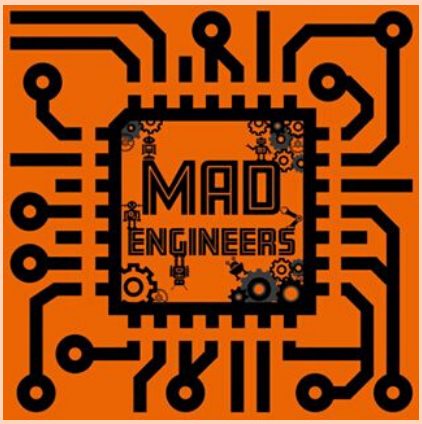
FTC TEAM #2123 MAD ENGINEERS

MANUFACTURING FEASIBILITY

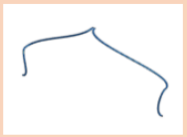
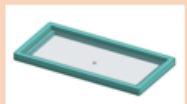
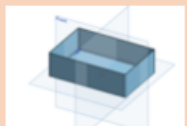


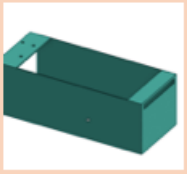
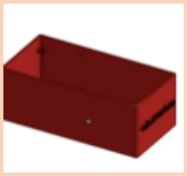
LOCATION Mumbai, India

SCHOOL On My Own Technology

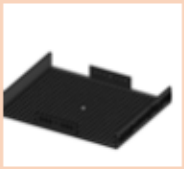
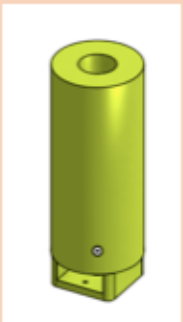
FIRST TECH CHALLENGE

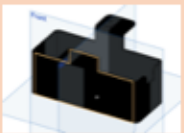
Onshape

Part name	Material	Weight (in g, rounded to 2dp)	Method of manufacturing	Cost of material, (in rupees)	Diagram
Box	polycarbonate	151.05g	Laser Cutting	7Rs	
Keypad Sanitizer Arm	aluminum 7050	149.69g	Laser Cutting	97Rs	
Pipes	plastic	5.44g	Laser Cutting	3Rs	
Credit Card Machine	polycarbonate	881.33g	Laser Cutting	40Rs	
Short L Beam	aluminum 7050	35.83g	Laser Cutting	23Rs	
Billing Machine	polycarbonate	449.06g	Laser Cutting	20Rs	
Packing Box Lid	polycarbonate	403.70g	Laser Cutting	18Rs	
Credit Card Machine Stand	aluminum 7050	95.71g	Laser Cutting	62Rs	
Bill Machine Cover	aluminum 7050	123.44g	Laser Cutting	80Rs	

Part name	Material	Weight (in g, rounded to 2dp)	Method of manufacturing	Cost of material, (in rupees)	Diagram
Long L Beam	aluminum 7050	72.12g	Laser Cutting	47Rs	
Cash sanitizer cover	aluminum 7050	403.18g	Laser Cutting	262Rs	
Side roller	polycarbonate	35.83g	Laser Cutting	2Rs	
Credit card machine holder	aluminum 7050	163.75g	Laser Cutting	106Rs	
Packing mechanism box	aluminum 7050	2750.58g	Laser Cutting	1787Rs	
Slider	aluminum 7050	73.94g	Laser Cutting	48Rs	
Base slider	aluminum 7050	196.41g	Laser Cutting	128Rs	
Outside chassis	aluminum 7050	374.67g	Laser Cutting	244Rs	
Outside chassis	aluminum 7050	374.67g	Laser Cutting	243Rs	

Part name	Material	Weight (in g, rounded to 2dp)	Method of manufacturing	Cost of material, (in rupees)	Diagram
Chassis churro	aluminum 7050	9.53g	Laser Cutting	Rs6Rs	
Chassis plate	aluminum 7050	33.31g	Laser Cutting	22Rs	
Motor side bush	aluminum 7050	3.07g	Laser Cutting	2Rs	
Large churro	aluminum 7050	0.18g	Laser Cutting	1Rs	
Belt wheel	polycarbonate	82.43g	Laser Cutting	36.9Rs	

Part name	Material	Weight (in g, rounded to 2dp)	Method of manufacturing	Cost of material, (in rupees)	Diagram
Small axle	aluminum 7050	0.91g	Laser Cutting	1Rs	
Base plate	aluminum 7050	2406.76g	Laser Cutting	1564Rs	
Axle	aluminum 7050	2.1g	Laser Cutting	1Rs	
Roller	polycarbonate	53.07g	Laser Cutting	3Rs	

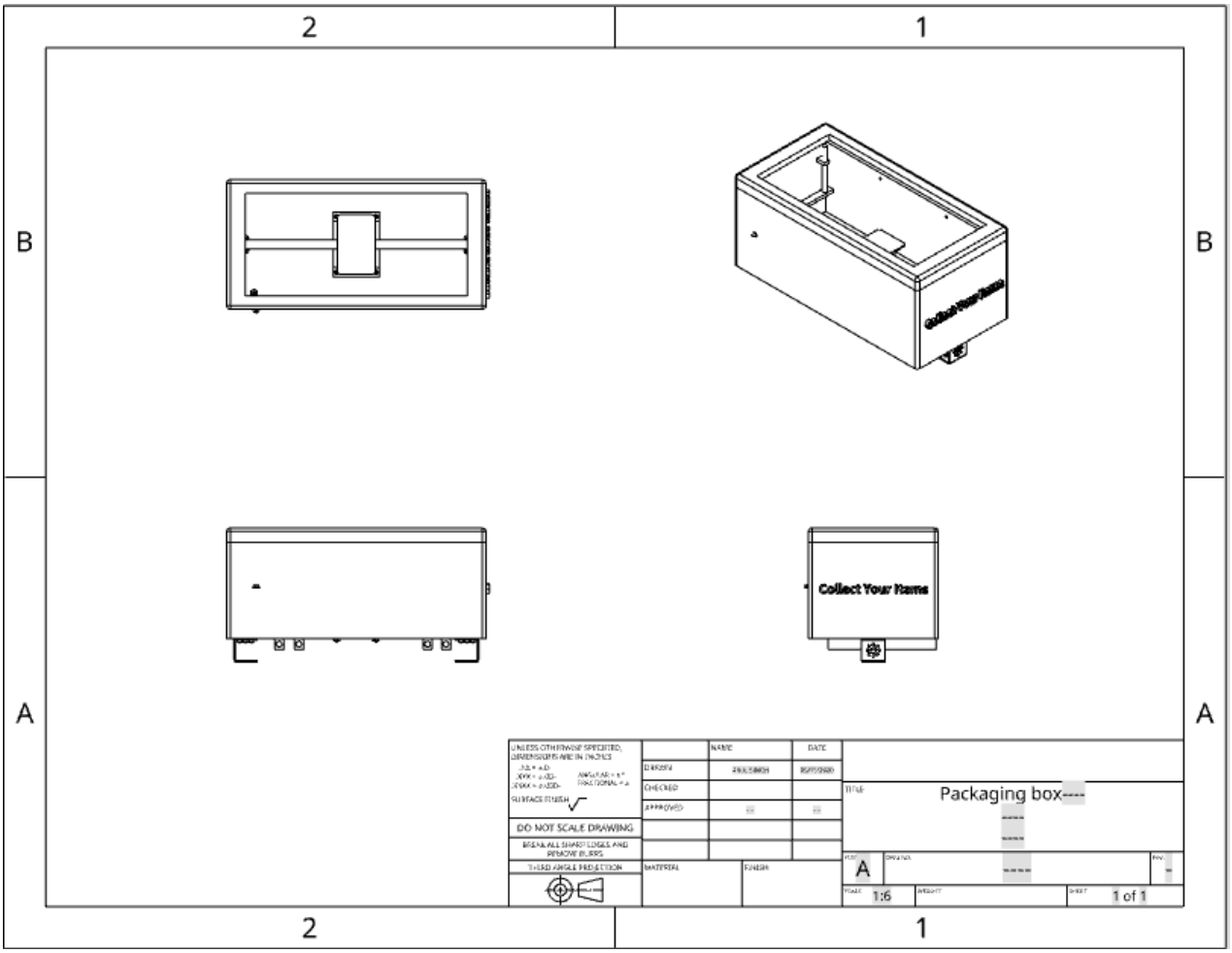
Part name	Material	Weight (in g, rounded to 2dp)	Method of manufacturing	Cost of material, (in rupees)	Diagram
Table roller	polycarbonate	586.50g	Laser Cutting	23Rs	
Phone Stand	aluminum 7050	55.35g	Laser Cutting	36Rs	
Phone Case	aluminum 7050	132.90g	Laser Cutting	86Rs	
Battery Case	polycarbonate	53.98g	Laser Cutting	24Rs	

Manufacturing note

All of the mechanical parts except the rollers are made in **aluminium**.This ensures the strength and lightness for the parts, thus leading to reduced weight and more effectiveness.The other parts that are more often used are made of **polycarbonate** because of its long lasting durability and also because of its ability to stay waterproof.

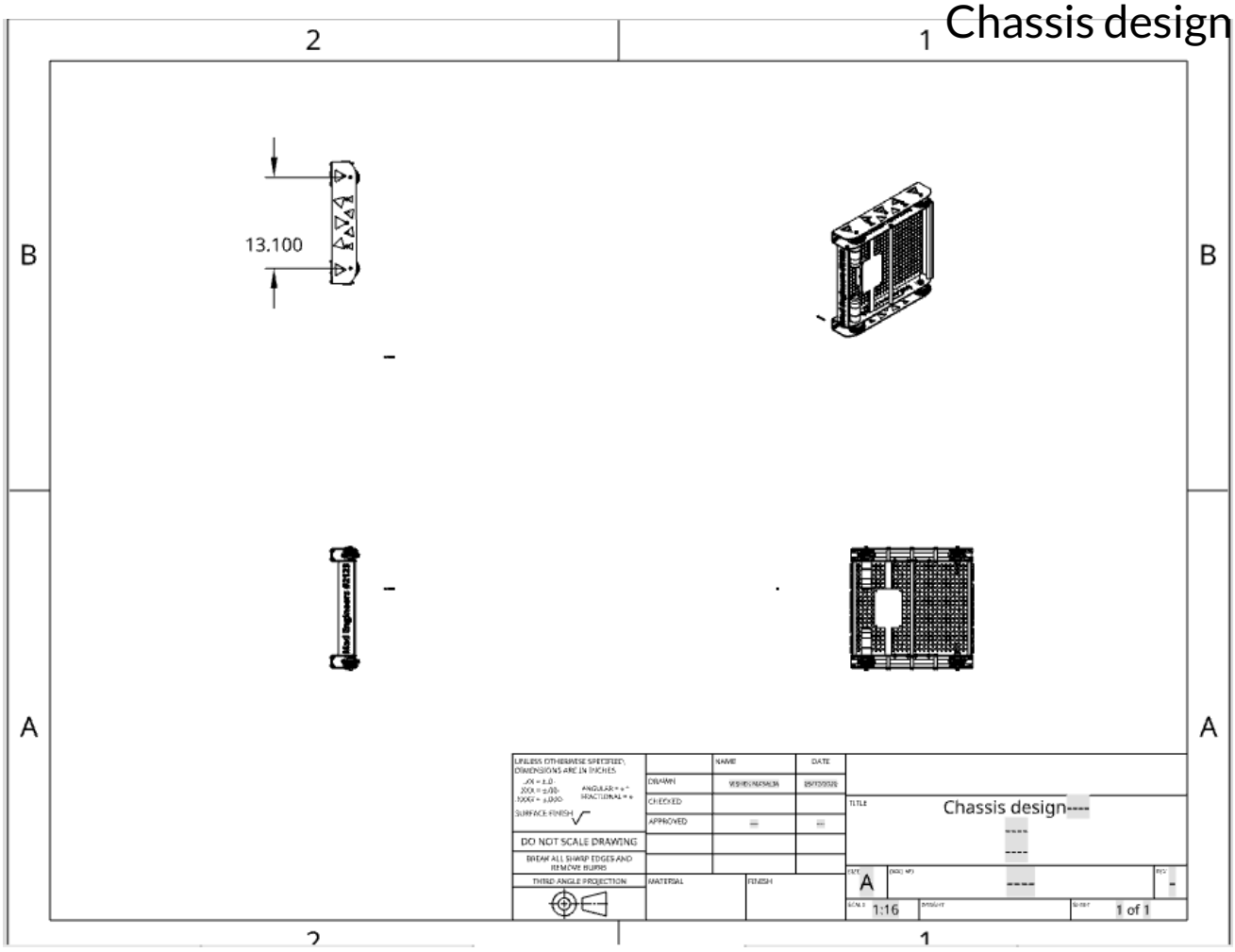
All of the parts are made using **laser cutting**, as it is very precise and effective.This way we ensure no problems in the manufacturing of the parts.

Mechanism drawings

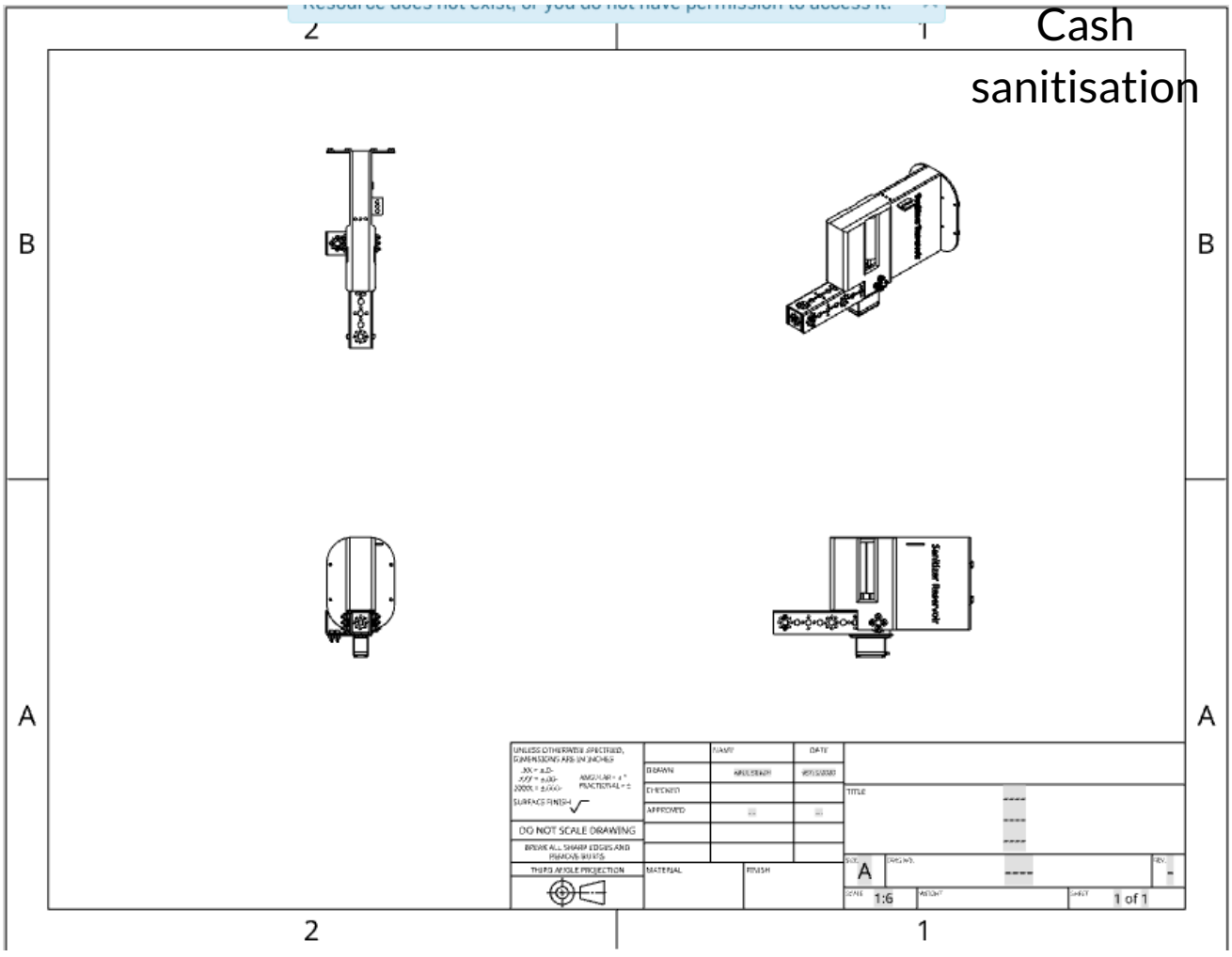


Packaging box

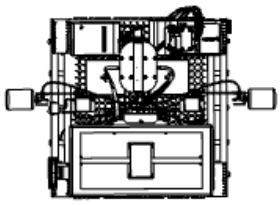
Chassis design

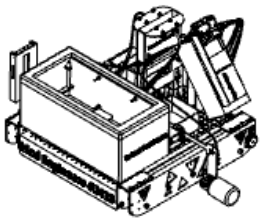


Cash sanitisation



Full robot









UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES		NAME	DATE
.XX = ±.03 ANGULAR = ± °	DRAWN	ARUL SURESH	09/07/2023
.XXX = ±.005 PRECISION = ±	CHECKED		
SURFACE FINISH: √	APPROVED	---	---
DO NOT SCALE DRAWING			
BREAK ALL SHARP EDGES AND REMOVE BURRS			
THIRD ANGLE PROJECTION	MATERIAL	FINISH	

TITLE: ---- ---- ----		
SIZE: A	DWG NO: ----	REV: --
SCALE: 1:12	WEIGHT:	SHEET: 1 of 1